Sheet 5.

1.a)

$$D_{KL}[Q(z|X) \,\|\, P(z|X)] = \sum_z Q(z|x) \log \frac{Q(z|X)}{P(z|X)}$$

$$= \sum_z Q(z|x) \left[\log Q(z|X) - \log P(z|X) \right]$$

$$= E_{z\sim Q(z|X)} \left[\log Q(z|X) - \log P(z|x) \right]$$

$$\overset{\text{Bayes}}{=} E_{z\sim Q(z|X)} \left[\log Q(z|X) - \log P(X|z) - \overset{\log}{P}(z) + \overset{\log}{P}(X) \right]$$

$$= \boxed{E_{z\sim Q(z|X)} \left[\log Q(z|X) - \log P(X|z) - P(z) \right] + \overset{\log}{P}(X)}$$

$$P(z|X) = \frac{P(X|z)P(z)}{P(X)}$$

1.b)

$$\log P(X) - D_{KL}[Q(z|X) \,\|\, P(z|X)] =$$

$$= - E_{z\sim Q(z|X)} \left[\log Q(z|X) - \log P(X|z) - \log P(z) \right]$$

$$= E_{z\sim Q(z|X)} [P(X|z)]$$

$$+ E_{z\sim Q(z|X)} \left[\log P(z) - \log Q(z|X) \right]$$

$$= E_{z\sim Q(z|X)} [P(X|z)] - D_{KL}[Q(z|X) \,\|\, P(z)]$$